

Ultimate Nutrition

Glucosamine Chondroitin

Glucosamine and chondroitin sulfate are substances found naturally in the body. Glucosamine is a form of amino sugar that is believed to play a role in cartilage formation and repair. Chondroitin sulfate is part of a large protein molecule (proteoglycan) that gives cartilage elasticity.

Increased activity/exercise, damages to the body and degenerative diseases may overcome the rate at which they are produced and used affecting the production of proteoglycans. Also, as we get older, the production of glucosamine and chondroitin sulfate by the body tapers. This in turn decreases synovial fluid that cushions and prevents the joints from rubbing together. Cartilage can begin to deteriorate and weaken (less collagen) producing damage, stiffness and pain.

Osteoarthritis is associated with a breakdown of cartilage in joints and commonly occurs in the hips, knees and spine. Osteoarthritis occurs when there is significant wearing of the articular cartilage of the joint. Articular cartilage is the cushioning substance on the ends of bones and consists of glycosaminoglycans. This substance is very viscoelastic and works well as a cushion. As the articular cartilage is worn down, swelling and pain occur due to inflammation.

Chondroitin and glucosamine, as dietary supplements, are used together to help ease the symptoms of osteoarthritis. Glucosamine is an amino sugar normally produced in the human body. It is necessary for the construction and repair of connective tissue and healthy joint cartilage. Cartilage is the protein part of the fibrous substance that holds the joints. It is also used in the formation of tendons, membranes, bones and other tissues.

Glucosamine is a major building block of proteoglycans and is needed to make glycosaminoglycans: proteins that bind water in the cartilage matrix. Each person produces a certain amount of glucosamine but age affects the body's ability to produce enough. An insufficient supply of glucosamine restricts the efficiency of cartilage rebuilding and can delay repair of cartilage and other connective tissues. Regular intake of glucosamine has been shown to slow Osteoarthritis progression by halting joint space narrowing and so maintain the health and mobility of the affected joints.

Chondroitin sulfate also occurs naturally in the body. Like glucosamine, chondroitin is a building block of proteoglycans: large protein molecules that give cartilage elasticity. Chondroitin sulfate acts by attracting fluid into the proteoglycan molecules. This synovial fluid also acts as a shock absorber and lubricates the joints. Chondroitin sulfate also stimulates collagen production, an ingredient that forms part of the cushioning structure at the ends of the joints.

Oral administration of chondroitin also prompts a significant increase in hyaluronic acid, a major component of synovial fluid: the lubricating fluid in joints. Like glucosamine,

chondroitin is a natural amino sugar already present in the body, but not always in sufficient quantity for joint health and repair.

SELECTED REFERENCES:

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FREQUENTLY ASKED QUESTIONS:

1: Who should take Glucosamine Chondroitin?

It is recommended for anyone with uncomfortable, stiff and aching joints. It can also benefit those who participate in regular physical activity by helping to maintain joint health and reduce stress from repetitive or jarring motions.

2: Is there anyone who shouldn't take this product?

If you are allergic to shellfish or fish, consult your doctor before deciding to take Glucosamine Chondroitin.

3: How long will it take for me to feel positive results?

The benefits of Glucosamine & Chondroitin supplementation develop over time. Generally, it may take 2 to 8 weeks before you see noticeable results.